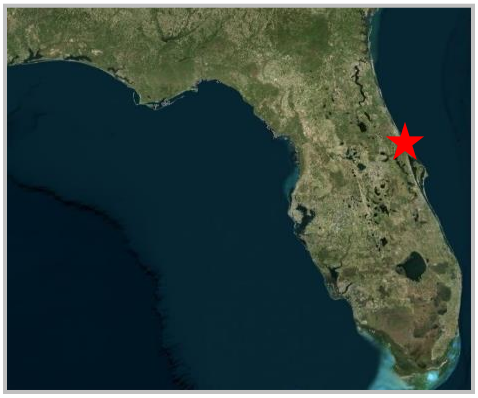
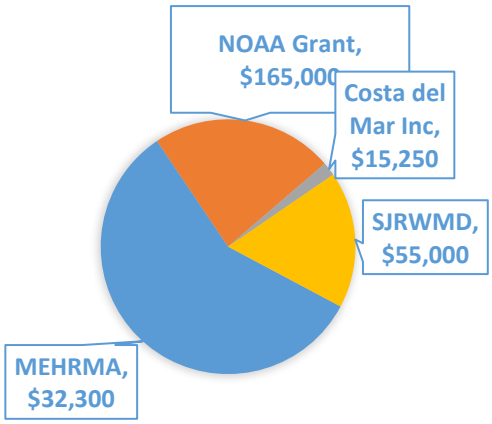
Installation of Native Vegetation at FWC's Mosquito Lagoon Marine Enhancement Center (New Smyrna Beach, Volusia County)





Partners

- St. Johns River Water Management District (SJRWMD) and Costa del Mar Inc. provided cash and staff time.
- Marine Discovery Center Inc. (MDC) and Brevard Zoo (BZ) provided volunteer time
- City of New Smyrna Beach provided coquina rip-rap and 3 days of spoil pile contouring
- USFWS provided cash for educational signs

Grant support

NOAA's Coastal Habitat Restoration program provided $165,000 toward the project.

Project Description

This project is a supplement to a partner-based and NOAA grant funded restoration of saltmarsh on FWC property in New Smyrna Beach. The 5-acre restoration was conducted between July and November, 2014, with SJRWMD contractors conducting the excavation of ~45,000 cubic yards of fill. During August-November, 2014, native plants (roughly 17,000 plants; various brackish wetland species) were purchased using MEHRMA funds for the project at cost of $9052. The plants were installed by FWC staff and partners, mainly volunteers associated with MDC, the non-profit organization housed at the FWC property. Additional plants and supplies were provided as match and a total of 25,000 plants were installed at the site. In addition, MEHRMA funds were used to purchase sod to secure the excavated sand storage mountain created as a result of the project and to purchase French drain supplies to repair eroding areas along a shoreline stabilization demonstration area created as part of the project. Excavated spoil was used to create a publicly accessible mountain overlooking the created saltmarsh, a future building pad, and an amphitheater adjacent to the marsh.

Objectives

The purpose of this project was to restore 5 acres of saltmarsh filled during the 1960s for development purposes (construction of a high school). The restored marsh serves to supplement

local fisheries via production of flora and fauna. It also serves regional restoration needs as propagated plants will be used for restoration efforts in the future. The marsh is used by students of all ages to learn monitoring and research techniques. Visitors to the MDC learn about saltmarsh ecology and the benefits of using environmentally-friendly shoreline stabilization techniques in the living shoreline demonstration area created as part of the project. Hiking trails on the site provide visitors with access to various Florida habitats and a vantage point from the overlook mountain. The shoreline and kayak launch offer opportunities for fishing and wildlife viewing.

Benefits

- Provides a vista for the local community to enjoy including hiking trails and shore fishing
- Propagates plants used for regional restoration projects
- Provides programs for students regarding monitoring and research techniques
- Provides education for visitors regarding sound shoreline stabilization techniques
- Provides fishing from shore and a kayak launch
- Provides nursery and foraging habitat for a variety of wildlife including juvenile fish, invertebrates, oysters, wading birds, shorebirds, and manatees

FWC Strategic Plan Elements

This project directly addresses the following FWC Strategic Plan Elements:

Theme I, Goal 2: Ensure sufficient habitats exist to support healthy and diverse fish and wildlife populations.

Strategies:

2. *Protect lands and waters critical to sustaining healthy and diverse fish and wildlife populations through diverse incentive programs.*
3. *Manage habitats to sustain healthy and diverse fish and wildlife populations.*

Theme 2, Goal 1: Ensure that Florida's fish and wildlife populations are able to sustain hunting, fishing and wildlife-viewing for current and future citizens and visitors.

Strategies:

2. Manage fish and wildlife populations to provide sustainable fishing, hunting and wildlife-viewing opportunities.

Acreage restored by habitat type (footprint)

- 5 acres of saltmarsh habitat restored to:
  - Harvest up to 55,000 plants annually for restoration projects
  - Provide 250 linear feet of shoreline stabilization demonstration area
  - Provide >1000 linear feet of shore for fishing

Acreage affected by restoration activities

Total project acreage of the saltmarsh including slopes is 5 acres whereas the mountain, future building pad, and amphitheater are 4 acres

Project Map (identify acreage of both footprint and affected area)





NSB project shown as before/after (June 2014; July 2015); altitude 400ft; saltmarsh and slopes 5 acres; mountain, future building pad, and amphitheater 4 acres

Project Photos



Photo 1: Volunteers plant native species at the restored marsh in NSB



Photo 2: Photopoint 4-3W (left to right) in June 2014 pre-construction, Oct 2014 nearing end of construction, and April 2015 ~5months post-construction.



Photo 3: Seawall retrofits at shoreline demo area in Nov 2014 (left) and Apr 2015 (right)



Photo 4: Mid-terrace planter at shoreline demo in Nov 2014 (left) and Apr 2015 (right)



Photo 5: Mountain erosion following heavy rains in Sep 2014 (left) and installation of common Bermuda



Photo 6: Installation of French drain in eroding pockets of shoreline demo area



Photo 7: Shoreline demo area (left to right) of native plants, terracing, kayak launch, retaining wall, seawall, and rip-rap



Photo 8: Random shots of plants funded through MEHRMA for the NSB project